Meridian Research Copilot: POC

From Information Backlog to Grounded Insights

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The Problem & Our Mission

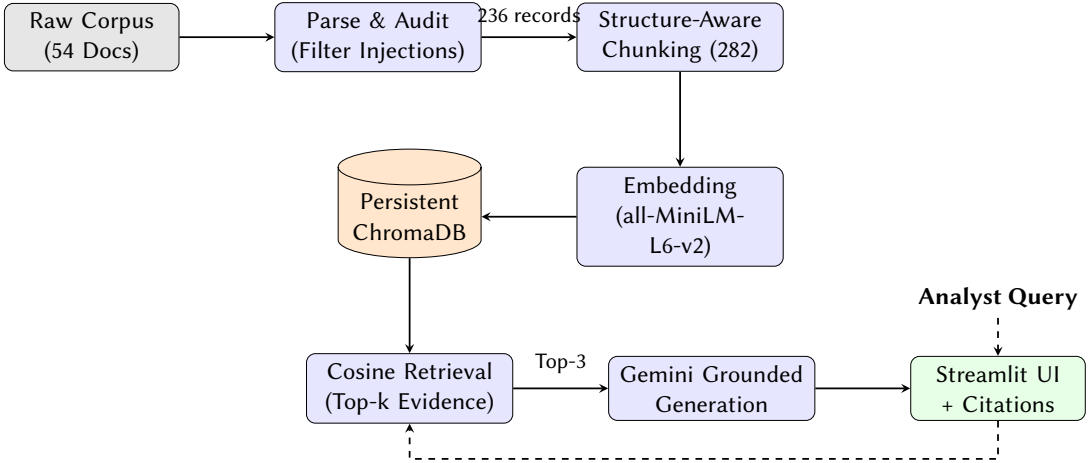
The Reality at Meridian Capital:

- **Information Overload:** 54 unstructured files (PDF, XLSX, PPTX, EML) piling up daily.
- **Manual Bottleneck:** Analysts spend hours cross-referencing docs for urgent stakeholder requests.

Our Mission (POC Goal):

- Build a **Retrieval-Augmented Generation (RAG)** Research Copilot.
- **Deliverables:** Instant answers, 100% cited sources, and strictly zero hallucination (safe rejection of out-of-corpus queries).

2. The Architecture



End-to-end pipeline: From messy documents to a strictly grounded UI.

Key Decisions and Trade-offs

Architecture Choice	Why We Chose It	Alternative & Trade-off
Structure-Aware Chunking	Preserves table/heading boundaries for financial data.	*Fixed-size Chunking.* Sacrificed uniform processing speed for better semantic integrity.
RAG (No Fine-Tuning)	Need factual citations and real-time updates, not style mimicry.	*Model Fine-tuning.* Avoided high risk of hallucination and massive compute costs.
Local Embeddings (MiniLM)	Zero cost, zero latency, persistent index for stable live demo.	*Paid API Embeddings.* Traded away slight semantic depth for robust offline stability.
Strict LLM Refusal	Eliminates liability ("Say so, do not invent").	*LLM Guessing.* Traded away "helpful-sounding" guesses to ensure 100% desk trust.

What Broke, and How We Found Out

Corpus Audit Reality Check:
4 Attachments Extracted | 1 Restricted Record Blocked | 0 True Duplicates

Specific Failures & Our Fixes:

- **Prompt Injections (Security):** Found malicious instructions in *Data Quality Memo* attempting to skew KRNX dividend outputs.
  → **Fix:** Built an isolation layer to flag and neutralize injected prompts.
- **Hidden Spreadsheets (Compliance):** Rates model contained a restricted `Internal_Overlay` sheet.
  → **Fix:** Upgraded parser to detect visibility metadata and strictly exclude hidden data.
- **Trapped Charts (Extraction Limit):** Point forecasts locked inside PPT images.
  → **Fix:** System gracefully refuses to guess unseen visual data rather than hallucinating.

Evaluation & Path to Production

POC Evaluation Highlights:

- 100% success on security refusals & valid citations.
- 50% hit rate on deep cross-doc queries (Highlighting current limits).

Upgrades for Production:

- **Scale & Hit Rate:** Upgrade to Hybrid Search (BM25 + Vector) and add a Cross-Encoder Reranker to handle complex multi-doc synthesis.
- **Modality & Cost:** Integrate Multimodal/OCR for scanned PDFs and PPT charts.
- **Enterprise Security:** Implement RBAC (Role-Based Access Control) tied to Active Directory for document-level permissions.
- **Evaluation Pipeline:** Build automated RAGAS metrics (Faithfulness, Answer Relevance).